

Pregnancy Risk Prediction

Model Comparison Report

Project Scope

Compare 4 classifiers for pregnancy risk levels:
LOW, MEDIUM, HIGH

Dataset

3600 synthetic rows
Balanced classes
Noise + overlap + medical interactions

Compared Models

- Random Forest
- Logistic Regression
- SVM
- XGBoost

Evaluation

- Synthetic split: 70/30 (stratified)
- Real-patient Excel validation
- Accuracy, F1 per class, training time

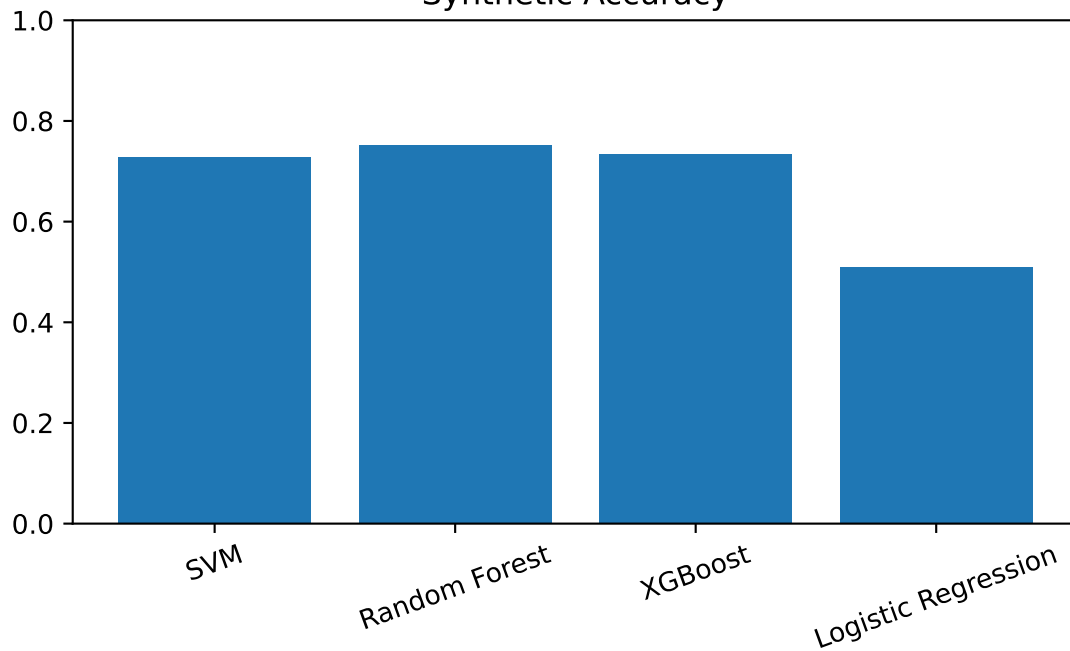
Side-by-Side Model Comparison

Model	Train Time(s)	Syn Acc	Real Acc	F1 HIGH	F1 LOW	F1 MEDIUM
SVM	0.149	0.728	0.867	0.828	0.766	0.582
Random Forest	0.529	0.752	0.800	0.856	0.779	0.608
XGBoost	0.686	0.733	0.800	0.848	0.762	0.578
Logistic Regression	0.026	0.510	0.467	0.691	0.504	0.277

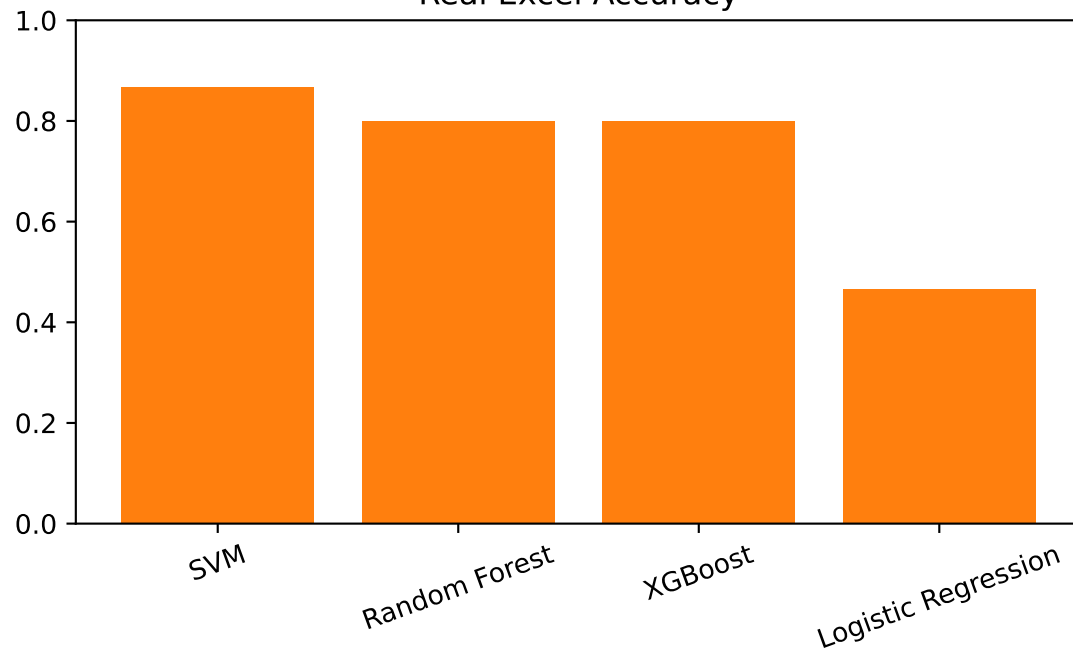
Best Real Validation Accuracy: SVM

Performance Charts

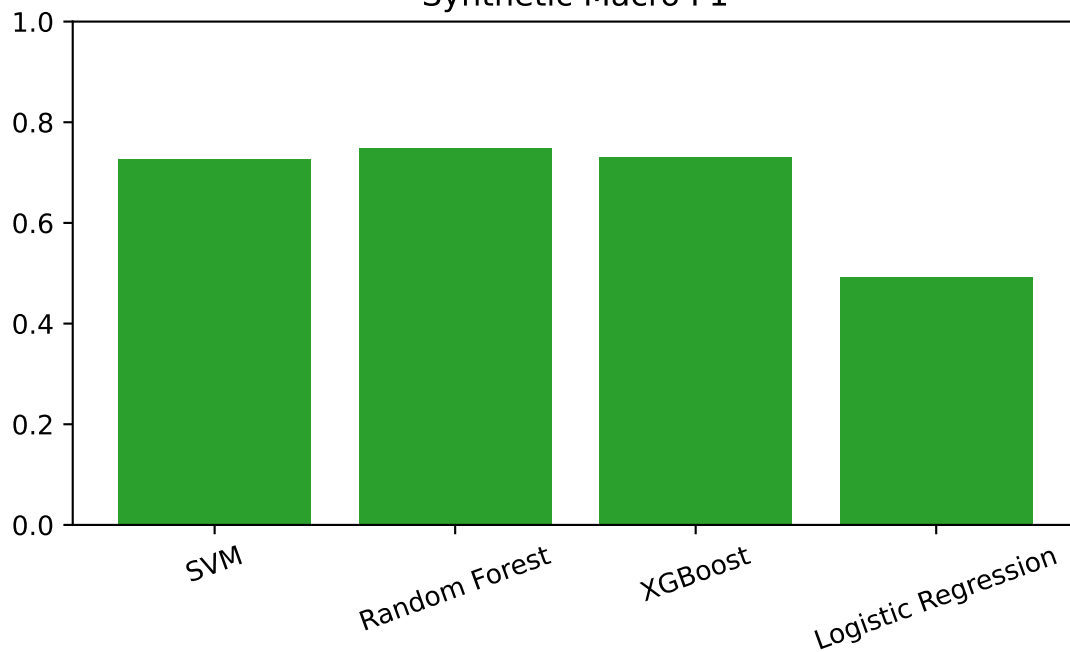
Synthetic Accuracy



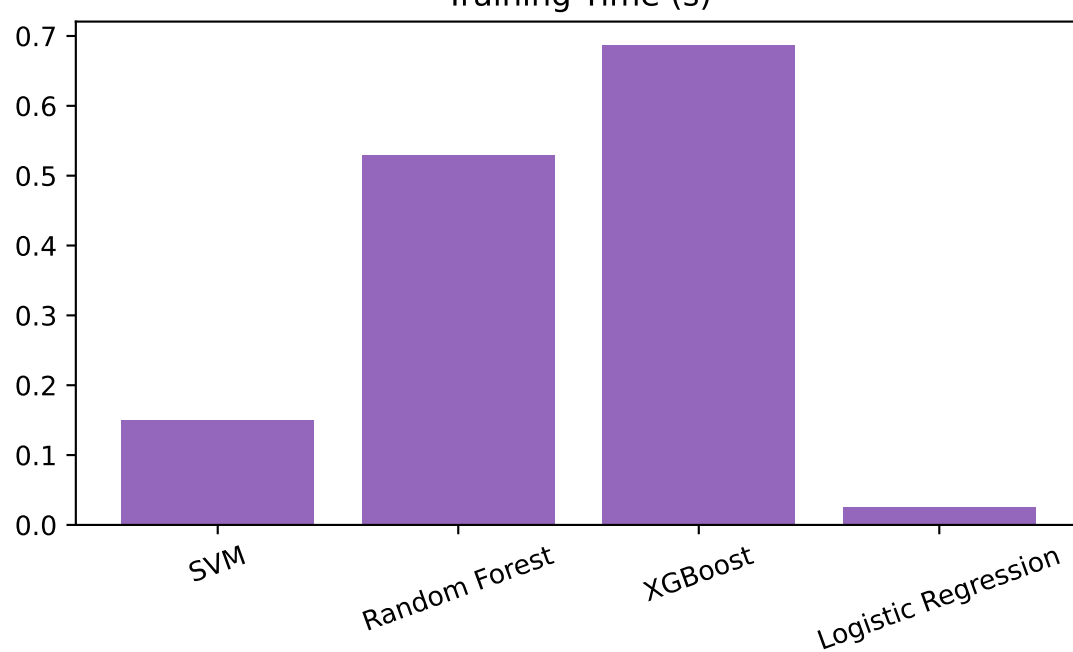
Real Excel Accuracy



Synthetic Macro F1



Training Time (s)



Model-by-Model Summary

SVM

Train Time: 0.149s | Synthetic Acc: 0.728 | Real Acc: 0.867

Synthetic F1 -> HIGH: 0.828, LOW: 0.766, MEDIUM: 0.582

Real F1 -> HIGH: 0.667, LOW: 0.947, MEDIUM: 0.750

Random Forest

Train Time: 0.529s | Synthetic Acc: 0.752 | Real Acc: 0.800

Synthetic F1 -> HIGH: 0.856, LOW: 0.779, MEDIUM: 0.608

Real F1 -> HIGH: 0.667, LOW: 0.900, MEDIUM: 0.571

XGBoost

Train Time: 0.686s | Synthetic Acc: 0.733 | Real Acc: 0.800

Synthetic F1 -> HIGH: 0.848, LOW: 0.762, MEDIUM: 0.578

Real F1 -> HIGH: 0.667, LOW: 0.900, MEDIUM: 0.571

Logistic Regression

Train Time: 0.026s | Synthetic Acc: 0.510 | Real Acc: 0.467

Synthetic F1 -> HIGH: 0.691, LOW: 0.504, MEDIUM: 0.277

Real F1 -> HIGH: 0.000, LOW: 0.667, MEDIUM: 0.333

Final Recommendation

Conclusion

Best model on real-patient Excel validation: SVM.

Random Forest remains a strong deployment choice in this PFE context because it provides a practical balance:

- Strong and stable performance across classes
- Simple workflow with little parameter tuning
- Robustness to noisy and overlapping medical patterns
- Reliable behavior on small and medium datasets

This makes Random Forest a safe baseline model for student-level clinical risk prototyping.